

Siddesh Patil

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SUMMARY

AI/ML practitioner with hands-on experience building end-to-end intelligent systems, including a solar-powered AIoT platform, a hybrid retrieval-augmented generation (RAG) pipeline, and a semantic search-based recommendation engine. Hands-on experience with the ML lifecycle across academic projects, from data preprocessing through model evaluation, with growing exposure to AWS cloud. Seeking roles in AI/ML engineering or data science.

PROJECTS

SoilStream | *Python, scikit-learn, TensorFlow, ResNet50, ESP32, Flask, Firebase* | [GitHub](#)

- Built an SVR-based rainfall prediction model achieving **92.86% accuracy and 0.93 precision** on Konkan regional climate data, outperforming LightGBM and GRU baselines.
- Trained a ResNet50-based CNN for multi-class crop identification achieving 97.2% test accuracy integrated with a rule-based irrigation controller to reduce waterlogging risk during monsoon season.
- Designed a solar-powered IoT sensing pipeline (ESP32 + BME280 + Firebase) with a mobile app for real-time field monitoring, pump control, and crop disease diagnosis via the Kindwise API.

VeriDoc | *Python, FAISS, BM25, Sentence-Transformers, Mistral, Gradio* | [GitHub](#)

- Built a hybrid retrieval pipeline over local PDFs combining dense retrieval (FAISS embeddings) and sparse keyword retrieval (BM25), merging results into a single ranked list before generating cited answers with an LLM (Mistral).
- Evaluated pipeline outputs using the RAGAS framework on synthetic question-answer pairs, tracking faithfulness and relevancy metrics across experiment runs with MLflow.

Sidekik | *Flask, Python, BERT, FAISS, Elasticsearch, Supabase, PostgreSQL* | [GitHub](#)

- Engineered a BERT + FAISS-based semantic matching engine for freelancer-project recommendations, using sentence embeddings and vector similarity search to rank candidates over a corpus of freelancer profiles and project records.
- Implemented semantic search with Elasticsearch and integrated the Adzuna API for real-time skill-demand analytics.
- Built a full-stack platform with Flask APIs, a Supabase backend, including a marketplace for freelancers to list completed work.

EDUCATION

University of Mumbai

2022 – 2026

B.E. Computer Science and Engineering (Data Science) | **Honours in AI/ML**

TECHNICAL SKILLS

Languages: Python, SQL, Java, C

AI/ML: NLP, RAG, Recommendation Systems, SVR, CNNs, LSTM

Frameworks & Libraries: Flask, scikit-learn, Pandas, NumPy, Matplotlib, Sentence-Transformers, FAISS

Cloud & Tools: Git, GitHub, Firebase, Supabase, Linux

PUBLICATIONS & INTELLECTUAL PROPERTY

Research Paper – IEEE CONIT 2026 (Accepted)

Paper ID: 960

A Solar-Powered AIoT-Based System for Intelligent Soil Monitoring and Automated Irrigation Control

Patent Filed – Government of India

App. No: 202621042187

Smart Solar-Powered AIoT System for Precision Agriculture (SoilStream)

CERTIFICATIONS

Generative AI

Google Cloud Virtual Internship (AICTE)

Google AI/ML

Google Cloud Virtual Internship (AICTE)

Data Science Methodology

IBM Developer Skills Network